

TechSupport RAG

Multi-Document & Image Troubleshooting Assistant — Project Synopsis

Problem Statement

Technical troubleshooting information is scattered across manuals, installation guides, FAQs, and error screenshots. Users — especially non-technical ones — must manually search multiple sources to identify an error's meaning, cause, and fix. This makes troubleshooting slow and frustrating.

Proposed Solution

A Naive Retrieval-Augmented Generation (RAG) system combining a multi-document and image knowledge base. Users ask troubleshooting questions in plain language; the system retrieves relevant text and image-derived context and generates a concise, grounded answer via an LLM.

Example Use Case

Query: “My router's internet light is blinking red. What should I do?” → System retrieves context from the troubleshooting guide, a lights-reference image, and the user manual → LLM returns a structured fix (check WAN cable, restart router, verify connection).

Objectives

- Ingest multiple PDFs (manuals, guides, FAQs) as knowledge base
- Ingest PNG/JPG troubleshooting images
- Extract text from PDFs and split into chunks
- Extract meaningful content from images via a vision-capable LLM
- Generate embeddings for all content and store in a vector database
- Retrieve relevant chunks/images per user query and generate a grounded answer
- Gracefully inform the user when no relevant information exists

Technology Stack

Component	Tool
Language	Python
Document Processing	pypdf, RecursiveCharacterTextSplitter
Image Understanding	Vision-capable LLM
Embeddings	OpenAI text-embedding-3-small
Vector Database	ChromaDB
LLM	GPT-4o-mini
Framework	LangChain
UI (optional)	Streamlit

Sample Queries

“My laptop is not charging — what should I check?” • “What does this red indicator mean?” • “Which port do I use for an external monitor?” • “My laptop is overheating — what should I do?”

Expected Output Format

